higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

T1450**(E)**(A11)T

NATIONAL CERTIFICATE

QUANTITY SURVEYING N5

(2050015)

11 April 2019 (X-Paper)
09:00–13:00

REQUIREMENTS: **Dimension paper (BOE 8/12)**
Abstract paper (BOE 8/10)
Billing paper (BOE 8/11)

Candidates will require their own unmarked Standard System of Measuring Building work.

Calculators may be used.

This question paper consists of 5 pages and 2 addenda.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
QUANTITY SURVEYING N5
TIME: 4 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. ALL the work in SECTION A must be done on the appropriate paper.
4. ALL the work in SECTION B must be done in the ANSWER BOOK.
5. Number the answers according to the numbering system used in this question paper.
6. Do NOT use tippex. Use the quantity surveying method for rectifications.
7. Use only BLUE or BLACK ink.
8. Write neatly and legibly.

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SECTION A

On ADDENDUM A (attached) the floor plan and sectional view of a mountain hut are shown. The specifications for the different trades are as follows:

SPECIFICATIONS:

Earthworks: – Excavation is in ordinary earth

Concrete: – 15 MPa

Masonry:
– Ordinary bricks built in stretcher bond
– Facebrick externally
– 1:4 cement mortar

NOTE: Use dimension paper to answer QUESTIONS 1–3.

QUESTION 1

Calculate the following:

1.1	The length of the external perimeter of the hut	(4)
1.2	The centre line of the external 330 mm foundation wall	(3)
1.3	The centre line of the internal 330 mm foundation wall	(5)
1.4	The height of the risk of collapse	(2)
1.5	The height of the 330 mm foundation wall	(1)
		[15]

QUESTION 2

Use the dimensions calculated in QUESTION 1 to measure the following:

2.1	Excavate for surface trenches excluding water and carting away of surplus material	(3)
2.2	Passing for excavation of surface trenches	(4)
2.3	Risk of collapse in ALL the foundation trenches	(3)
2.4	Passing for risk of collapse	(4)
		[14]

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QUESTION 3

Measure the following:

3.1	330 mm thick foundation wall	(3)
3.2	Mass concrete in footings	(3)
3.3	Passing for concrete in footings	(3)
3.4	Backfilling in trenches	(3)
3.5	Passing for backfilling in trenches	(4)

[16]

QUESTION 4

Refer to the answers in QUESTION 2 and QUESTION 3 and do only the following:

4.1	Square only the earthwork.	(4)
4.2	Abstract ALL the earthwork items.	(5)
4.3	Bill ALL the earthwork items.	(6)

[15]

TOTAL SECTION A: 60

SECTION B

ALL the work in SECTION B must be done in the ANSWER BOOK.

QUESTION 5

Prepare the interim valuation for the fourth partial payment to the contractor.

ALL calculated amounts to be rounded off to the nearest 10 Rand.

A summary of the bill of quantities is shown on ADDENDUM B (attached).

- The contract period is 9 months.
- 100% of the earthwork is complete.
- 100% of the concrete work is complete.
- 85% of the masonry is complete.
- 10% of the waterproofing is complete.
- 15% of the carpentry and joinery is complete.
- The value of unfixed material on site is R6 500,00.
- A retention fee of 10% is withheld to a maximum of 5% contract value.
- The contractor was paid a total amount of R682 350,00 for the first three months. **[10]**

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QUESTION 6

6.1 State the principles laid down by the Standard System on how the following should be measured:

6.1.1 Stripping of topsoil (3)

6.1.2 Open-face excavations (4)

6.2 Explain the following terms:

6.2.1 Nominated suppliers

6.2.2 Nominated subcontractors

(2 × 4) (8)
[15]

QUESTION 7

When a building site has to be established, what important considerations have to be taken into account when planning to incorporate the following:

7.1 Access road

7.2 Plant and machines

7.3 Site offices

7.4 Storage areas

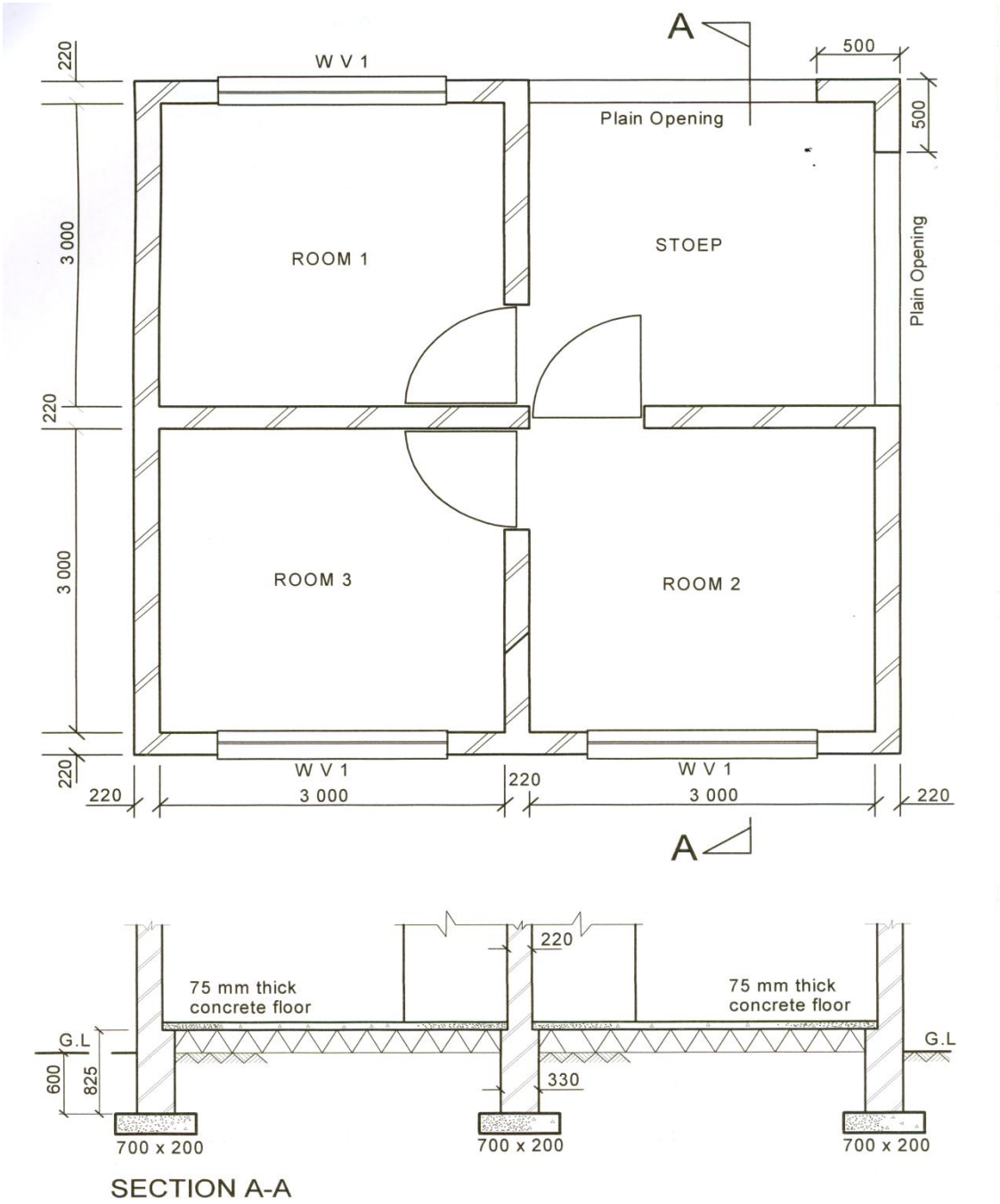
7.5 Clearing the site

(5 × 3) **[15]**

TOTAL SECTION B: 40
GRAND TOTAL: 100

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ADDENDUM A



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ADDENDUM B

BILL OF QUANTITIES

1.	Preliminary and general	R125 300,00
2.	Earthworks	R99 510,00
3.	Concrete, formwork and steel reinforcement	R457 678,00
4.	Masonry	R220 300,00
5.	Waterproofing	R17 453,00
6.	Roof coverings	R205 321,00
7.	Carpentry and joinery	R67 423,00
8.	Ceilings	R25 308,00
9.	Floor coverings	R41 523,00
10.	Ironmongery	R6 789,00
11.	Metalwork	R42 834,00
12.	Plastering	R20 318,00
13.	Plumbing and drainage	R45 756,00
14.	Glazing	R15 909,00
15.	Painting	R45 778,00
16.	Contingency sum	R62 155,00
		R1 492 375,00

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